Exp 3b: Analysis of the Behaviour of Network Devices (HUB and Switch) Using Cisco Packet Tracer

AIM :

To analyse the behaviour of network devices (HUB and Switch) using the Cisco Packet Tracer simulator by designing small networks of end devices, tracing the communication between them and finding out the functioning difference between a hub and a switch.

ALGORITHM :

1. Start the experiment.
2. From the network component box, drag and drop 4 generic PCs and one HUB, and another 4 generic PCs and one Switch, onto the workspace.
3. Using the copper straight-through cable (Connections), connect each PC to the HUB; the link LED glows green when the link is up. Similarly, connect the 4 PCs to the Switch.
4. Assign an IP address and subnet mask to each PC (Desktop → IP Configuration). The default gateway and DNS server are not needed because all the devices are in the same network.
5. Select the PDU (message icon) from the common tools bar, click the source PC and then the destination PC connected to the HUB.
6. Observe the flow of the PDU from the source PC to the destination PC in Realtime / Simulation mode.
7. Repeat steps 5 and 6 for the PCs connected to the Switch.
8. Observe how the HUB and the Switch forward the PDU, and write the observation and conclusion about their behaviour.
9. Stop the experiment.

PROCEDURE :

Network 1 – only a HUB (4 hosts): Place 4 generic PCs and one HUB on the workspace. Click Connections, select the copper straight-through cable, click one PC and then the HUB. Repeat for the remaining 3 PCs.

Network 2 – only a Switch (4 hosts): Place 4 generic PCs and one Switch on the workspace and connect the 4 PCs to the Switch with copper straight-through cables in the same way.

IP configuration: Click a PC, go to the Desktop tab, click IP Configuration, select Static and enter the IP address and the subnet mask.

Device	IP address	Subnet mask
PC0	10.1.1.1	255.0.0.0
PC1	10.1.1.2	255.0.0.0
PC2	10.1.1.3	255.0.0.0
PC3	10.1.1.4	255.0.0.0

Testing: Click the PDU icon in the common tools bar, drop it on the source PC and then on the destination PC. In the Realtime mode / Simulation mode observe how the PDU travels. Repeat the same test for both networks.

Observation:

- A **HUB** works at the physical layer. It does not know the destination; it simply repeats (broadcasts) the signal received on one port to all its other ports, so every connected PC receives the packet and only the intended PC accepts it.
- A **Switch** works at the data link layer. It learns the MAC addresses of the connected PCs and forwards the frame only to the port of the destination PC (it floods the frame to all ports only until the destination address is learnt), which reduces the unnecessary traffic.

OUTPUT :

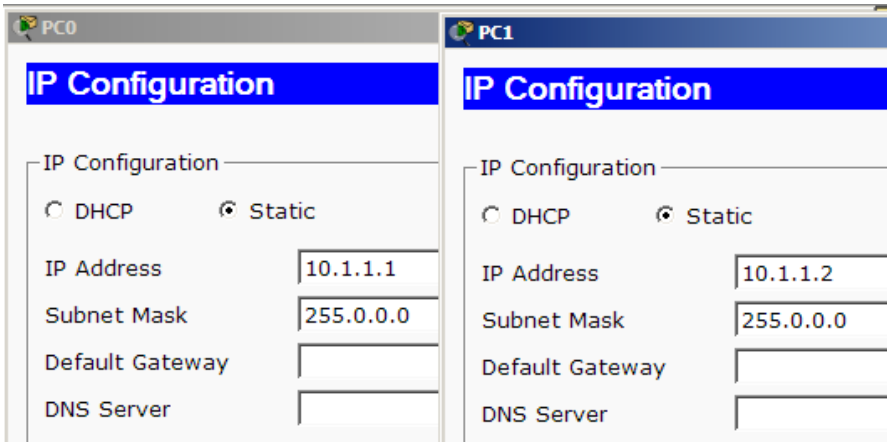


Fig 1. IP configuration (static) of PC0 and PC1 connected to the device

RESULT :

Thus, the behaviour of the HUB and the Switch was analysed using the Cisco Packet Tracer simulator. It was observed that a HUB forwards the received packet to all its ports, whereas a Switch forwards it only to the destination port.